bnjannuzzi@wm.edu | Website: https://bailey-jj.github.io/ | GitHub: https://github.com/Bailey-JJ

Bailey Jannuzzi

PhD Student, Mathematical and Computational Biology
Department of Applied Science
William & Mary

Education

William & Mary | Applied Science Department Williamsburg, VA
PhD in Mathematical and Computational Biology Expected: 2031
Advisor: Geoffrey Zahn, PhD

Utah Valley University Orem, UT
Bachelor of Science in Bioinformatics | Magna Cum Laude Spring 2026
GPA: 3.81

Research Experience

Utah Valley University Orem, UT
Hjelmen Lab | Independent Student Researcher Spring 2024 - Present
Research Advisor: Carl E. Hjelmen, PhD
Investigating the diet of forensically relevant blow flies (Diptera: Calliphoridae) to determine their effectiveness as wildlife monitors.

- Constructing reproducible bioinformatic pipelines to identify environmental DNA from blow fly gut content samples.

Utah Valley University Orem, UT
Hjelmen Lab | Peer Research Collaborator Spring 2025 - Present
Research Advisor: Carl E. Hjelmen, PhD
Assisting with gathering data on blow fly (Diptera: Calliphoridae) diversity and distribution in Utah.

- Conducting field collections.
- Preparing collected specimens through pinning and labeling to be identified.

Presentations

***Presenting author**

Poster Presentations

American Society for Microbiology, Intermountain Branch Meeting 2026
Evaluating Genomic Pipelines for Predicting Bacteriophage Host Range in Salmonella enteric, ***Rogers N, Jannuzzi B, Johnston M, Lopez A, Brooks L

Entomological Society of America, Pacific Branch Meeting 2025
Oxford Nanopore Technology Sequencing and Identification of Fly Gut Content, ***Jannuzzi B, Meeds A, Weidner LM, and Hjelmen CE

Entomological Society of America, Annual Meeting 2025
From City to Canyon: A Survey of Blow Fly Biodiversity Across Utah Ecoregions, ***Beck H, Jannuzzi B, Weidner LM, and Hjelmen CE

Oral Presentations

North American Forensic Entomology Association, Annual Meeting | 1st Place Award 2026
From Gut to Database: Leveraging Blow Fly Dietary DNA for Forensic and Conservation Applications, ***Jannuzzi B, Hornsby D, Meeds A, Weidner LM, and Hjelmen CE

North American Forensic Entomology Association, Annual Meeting 2026

Characterizing Calliphoridae Diversity in the Mountain West: A Statewide Survey of Blow Fly Communities in Utah, ***Beck H**, Jannuzzi B, Weidner LM, Hjelmen CE.

Entomological Society of America, Annual Meeting | 2nd Place Award 2025

Optimizing a Bioinformatic Pipeline for Profiling Blow Fly Diets, ***Jannuzzi B**, Meeds A, Weidner LM, and Hjelmen CE

Entomological Society of America, Annual Meeting 2025

What flies eat and excrete: Insights from puke, poop, and gut contents of blow flies for wildlife and forensic applications, ***Weidner LM**, Jannuzzi B, Yeilee Perez Ortiz, Gabrielle Meza, Meeds A, and Hjelmen CE

Awards and Funding

Funding (Totaling $3,719)

Undergraduate Research, Scholarly, and Creative Activity Dissemination Grant 2025

Amount Awarded: $1,093

Scholarly Activities Committee Grant 2025

Amount Awarded: $1198

Undergraduate Research, Scholarly, and Creative Activity Grant 2025

Amount Awarded: $1428

Awards and Honors

Outstanding Student Award for Biology — Utah Valley University, College of Science Spring 2026

Undergraduate 10-Minute Presentation: SysEB, Behavior and Biodiversity | 2nd Place Award 2025

Dean's List — Utah Valley University Spring 2023 - 2026

Experience

Utah Valley University Orem, UT

General Biology Teaching Assistant Fall 2025

- Grading assignments and exams.
- Hosting weekly study and review sessions.
- Assisting with in-class activities.
- Teaching vignette style lectures in-class.

Utah Valley University Orem, UT

Genetics Teaching Assistant Spring 2025

- Supporting students in applying class concepts to solve practical problems.

Membership

UVU Evolution and Bioinformatics Club | President Spring 2025 - 2026

UVU Evolution and Bioinformatics Club | Secretary Spring 2024 - 2025

American Society for Microbiology, Intermountain Branch Current

North American Forensic Entomology Association Current

Entomological Society of America Current

Entomological Society of America, Pacific Branch Spring 2025 - 2026

Skills

Computational Skills

R, Python, SQL, BASH Scripting/Linux, Bioinformatic Pipeline Development (Oxford Nanopore sequencing, eDNA metabarcoding)

Laboratory Skills
DNA Extraction, PCR, gel electrophoresis, DNA purification, insect specimen preparation (pinning & labeling)